

UQFF P14: CMB-S4 Spectral Distortion μ -Parameter Strict Bound from R26 Closure

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Abstract

The UQFF closed Lagrangian, with all 26 radial layers locked at saturated occupation, predicts vanishing additional CMB chemical-potential spectral distortion: $\mu_{\mathrm{UQFF}} \leq 1.0 \times 10^{-8}$. This sits at the projected 1σ floor of CMB-S4 (and just below LiteBIRD/PIXIE forecasts). Detection of $\mu > 3.0 \times 10^{-8}$ at 3σ falsifies the R26 saturation assumption. This is the cosmological-microwave-background counterpart of PAPER_1178 (DESI Y5 strict-static dark energy).

1. UQFF Prediction

The R26 manifold has zero free parameters in cosmic time at $z < 10^6$ (recombination + diffusion-damping era). Energy injection into the CMB photon bath is bounded above by the residual buoyancy ledger residual, which is geometrically fixed at

$$\mu_{\mathrm{UQFF}} = \frac{\rho_{\mathrm{BSFG}}}{\rho_{\gamma,0}} \cdot \xi_0 \cdot f_{\mathrm{damp}} \leq 1.0 \times 10^{-8}$$

where $\rho_{\mathrm{BSFG}} = 5.96 \times 10^{-10} \text{ J/m}^3$ (4-term G2 ledger), $\rho_{\gamma,0} = 4.17 \times 10^{-14} \text{ J/m}^3$, $\xi_0 = 13/3$, and the damping-era suppression $f_{\mathrm{damp}} = e^{-(z_{\mu} / 5 \times 10^4)^{5/2}}$ with $z_{\mu} \sim 2 \times 10^6$ saturates the integral to $\leq 10^{-8}$.

2. Observational Channels

Experiment	Year forecast	1σ on μ
FIRAS (1996)	published	9×10^{-5}
PIXIE forecast	late 2020s	1×10^{-8}
CMB-S4	2030-2033	1×10^{-8} (broad)
LiteBIRD	2032-2034	$\sim 1 \times 10^{-8}$

3. Falsification Criterion

UQFF is falsified if any of these experiments measure

$$\mu_{\mathrm{obs}} > 3.0 \times 10^{-8} \quad \text{at} \quad 3\sigma.$$

4. CP4 Implementation

```
python
from CondensedPhysics4 import UQFFCMBmuDistortionCalculator
r = UQFFCMBmuDistortionCalculator().compute()
# returns: mu_UQFF_upper_bound = 1.0e-8,
# mu_falsifier_3sigma = 3.0e-8,
# decisive_experiment = "CMB-S4 / LiteBIRD (2030-2034)"
```

5. References

- [1] PAPER_1174 — UQFF closed Lagrangian and 4-term G2 ledger.
- [2] PAPER_1178 — UQFF P13 DESI Y5 strict-static dark energy.
- [3] Chluba & Sunyaev 2012 — CMB spectral distortion forecasts.
- [4] CMB-S4 collaboration 2022 — Science Book.